

Gazipur Digital University

Department Of Software Engineering

Project Proposal

“ReVendo: Buy, Sell & Repair Pre-Owned Items”

Submitted By:

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1 Problem Analysis and Motivation

Many buyers and sellers in Bangladesh rely on informal methods such as social media groups for second-hand transactions. These approaches are not only inefficient but also risky due to lack of transparency. Sellers struggle to reach potential buyers, while buyers cannot verify product authenticity and condition, leading to misunderstandings and fraud possibilities. Many usable items are discarded instead of repaired or recycled, which shortens product lifecycles, increases electronic waste, and contributes to environmental contamination.

We propose **ReVendo** in front of you today as a digital platform designed to simplify second-hand transactions in Bangladesh addressing these issues. It provides a secure, automated, and user-friendly system where users can list, browse, purchase, and repair pre-owned items efficiently. The platform, by product, connects users with trusted local repair providers to lengthen product lifecycles, and ensure transparency and trust between buyers and sellers.

Furthermore, ReVendo promotes eco-friendly practices by tracking environmental impact and rewarding users with eco-scores that encourage reuse and reduce e-waste. The motivation is to connect buyers and sellers while promoting a circular ecosystem that minimizes waste, lengthens product's lifespan, and promotes sustainable consumption.

2 Literature Review

Re-commerce, or reverse commerce, refers to the resale of pre-owned goods through digital platforms [1]. Globally, the second-hand market is growing rapidly and is projected to reach USD 367 billion by 2029, reflecting rising demand for sustainable consumer practices [2]. In Bangladesh, e-commerce has expanded quickly since the introduction of online payments in 2009, with the B2C market valued at USD 6.5 billion in 2022 [3]. Platforms like Bikroy.com remain popular, but issues of trust, payment security, and product authenticity still discourage participation [4].

Research shows that extending product lifecycles through repair services can reduce environmental impact by up to 70% compared to purchasing new products [5]. At the same time, improved communication systems within marketplaces have been shown to increase transaction completion rates and reduce disputes significantly [6]. Donation and recycling features further strengthen circular economy initiatives, while eco-scoring and gamification increase user engagement and encourage sustainable consumption [7, 8].

3 Methodology

The ReVendo system follows a structured workflow to ensure smooth transactions and sustainability. Users first register or log in, then interact with the marketplace to buy, sell, donate, or send items for repair. The system also combines eco-impact tracking and communication features to enrich user experience. An overview of this methodology is pictured in Figure 1.

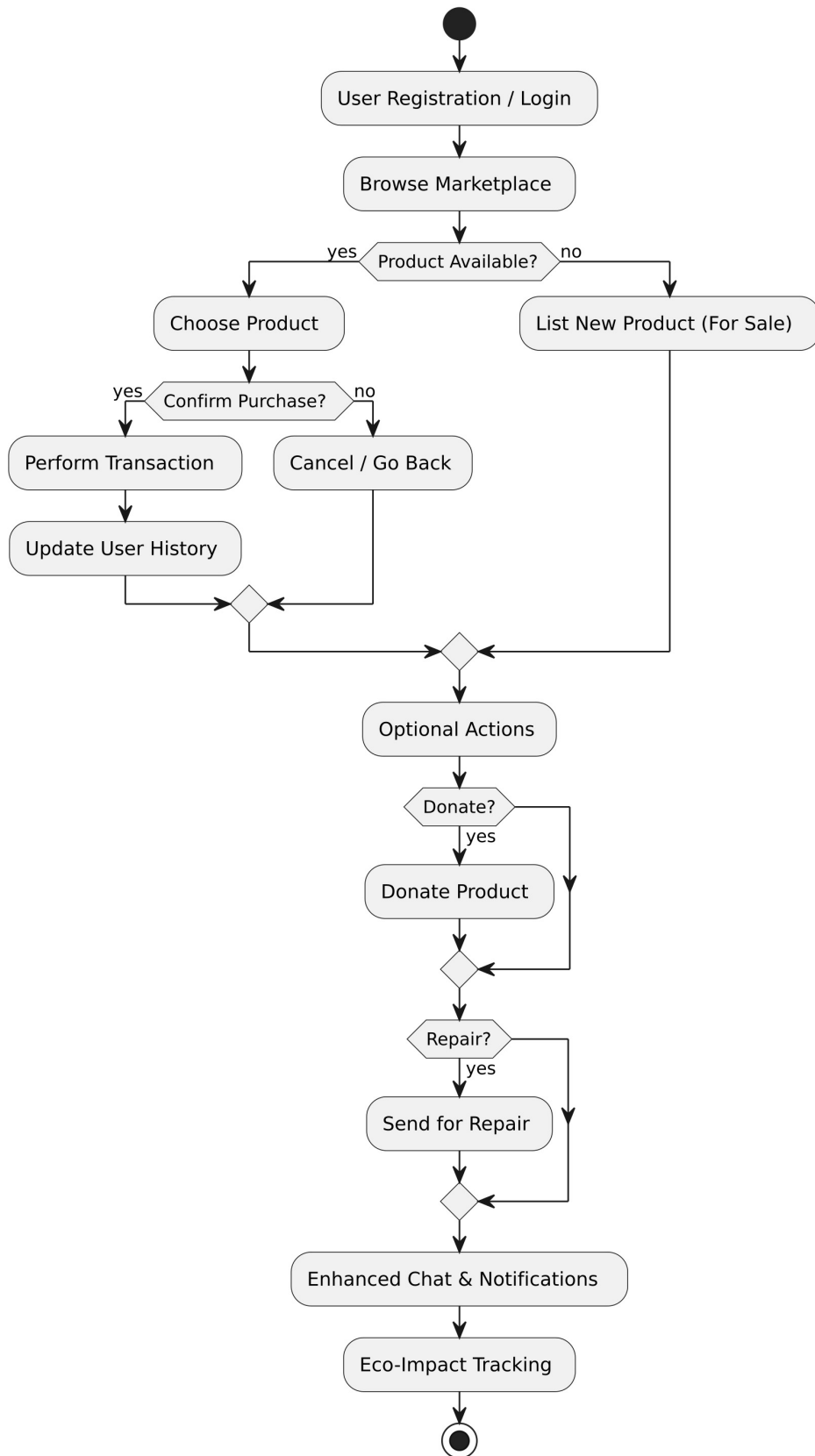


Figure 1: Block Diagram of The ReVendo System

4 Feasibility Study

The purpose of this study is to ensure that the proposed ReVendo system is economically, technically, and operationally viable.

4.1 Economic Feasibility

- Website/app development and server hosting
- Marketing and customer support costs
- Revenue from transaction fees and repair commissions

4.2 Technical Feasibility

- Secure handling of user data
- Fast search and product management system
- Real-time chat and eco-score tracking

4.3 Operational Feasibility

- Fraud prevention and secure payments
- User verification and trust building
- Repair service and donation coordination

4.4 Risk Analysis

- **Technical Risks:** Server downtime or security breaches
- **Operational Risks:** Fake accounts or fraudulent listings
- **Economic Risks:** High costs or low user adoption

Mitigation includes secure hosting, data encryption, user verification, and backup systems.

5 Main Phases of the Project

The project will be developed in the following main phases:

- Project proposal and planning
- Requirement specification of the project
- Selection of an appropriate SDLC model for the project
- Developing Data Flow Diagram (DFD) for the system

- Developing UML Use Case Diagram
- Developing UML Sequence and Communication Diagrams
- Developing UML Class Diagram
- Software testing

The detailed schedule for these tasks is shown in Table 1.

Sl.	Task	Week	Responsible	Phase
1	Requirement Specification and Data Collection	1	Emon & Sohan	Research & Planning
2	Requirement Finalization	1	Emon	Analysis
3	System Design and Modeling	1	Tasbih, Sohan & Emon	Design
4	System Modeling and Finalization	1	Emon & Tasbih	Design
5	System Development (Coding)	1	Emon, Sohan & Tasbih	Implementation
6	Testing and Feedback Sharing	1	Tasbih & Sohan	Testing
7	Beta Version Delivery for Feedback	1	Sohan, Tasbih & Emon	Testing
8	Feedback Sharing and Requirement Change	1	Sohan	Testing
9	Delivery Preparation with Documentation	1	Tasbih	Deployment

Table 1: Project Task Schedule for (ReVendo: Buy, Sell & Repair Pre-Owned Items)

6 Work Plan of the Project

Project Activities	Weeks									
	1	2	3	4	5	6	7	8	9	10
Planning										
Design										
Coding										
Testing										
Delivery										
Enhanced Features Integration										
User Training & Documentation										

Table 2: Project Development Timeline

7 Budget Details

Item	Description	Cost (TK)
Development Laptop	Existing device used for coding and design	0
Internet Connection	Monthly package for development and testing	1,000
Software Tools	Free and open source tools (IDE, GitHub, MySQL, Figma)	0
Marketing and Promotion	Posters, digital ads, social media boosts	5,000
Miscellaneous	Printing, diagrams, small project expenses	500
Total		6,500

Table 3: Estimated Budget Details

8 Conclusion

ReVendo is more than a marketplace—it is a circular economy platform that integrates buying, selling, repair, and donation to extend product lifecycles and reduce waste. With eco-impact tracking. It promotes sustainable consumption while increasing community engagement. This ranks ReVendo as a modern alternative to traditional e-commerce, addressing both environmental and social challenges.

References

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